MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department of Computer Science and Engineering
Course Hand-out

A. Basic Details:

Programme Name:	Bachelor of Technology
Course Name:	Android App Development
Course Code:	CSE3142
LTPC *(Lecture Tutorial Practical Credits):*	3 0 0 3
Session:	July – Nov, 2026
Class:	Semester – V
Course Coordinator:	Dr. Usha Jain
Course Instructor(s):	Dr. Usha Jain, Dr. Surbhi Sharma, Ms. Kirti Paliwal
Additional Practitioner(s) – if any *(Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):*	To be identified and appointed later

B. Introduction: This course is offered by the Department of Computer Science & Engineering to provide students with a comprehensive introduction to Android application development. Students will learn how to build modern, interactive mobile applications using Android Studio, explore the Android architecture, and understand essential concepts such as layouts, activities, data storage, and networking. The course blends theoretical concepts with practical experience, preparing students to design, develop, and deploy Android apps.

C. Course Outcomes:

CO Statement	CO	Level	Target Attainment %	Target Attainment level
CSE3142.1. Recall the architecture of the Android operating system and the key components of the Android development environment. [L1; Remember]	1	2 (Understand)	≥ 80	3
CSE3142.2. Explain the use of XML for UI design and demonstrate the implementation of various Android UI components and layouts. [L2; Understand]	2	3 (Apply)	≥ 70% < 80%	3
CSE3142.3. Develop Android applications using activities, intents, and fragments to create multi-screen applications and manage user interactions. [L3; Apply]	3	3 (Apply)	≥ 70	2
CSE3142.4. Build data persistence techniques such as Shared Preferences, SQLite databases. [L3; Apply]	4	3 (Apply)	≥ 70	2
CSE3142.5. Analyze and integrate advanced features such as networking APIs, background tasks, notifications, and sensors to enhance the functionality of Android applications. [L4; Analyze]	5	4 (Analyze)	≥ 60	1

Information about attainment levels:

Attainment (%)	Level
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO.1]. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

[PO.2]. **Problem analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using

first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4]. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. **Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6]. **The Engineers and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practices.

[PO.7]. **Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices.

[PO.9]. **Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. **Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12].**Life-long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSO1. Design and develop optimized computational solutions using appropriate programming paradigms, contributing to sustainable development to ensure efficient resource utilization.

PSO2. Create intelligent or innovative solutions to real-world challenges by applying modern tools, software engineering practices, and emerging technologies.

PSO3. Process and communicate data effectively while ensuring software security, data privacy, and reliable data communication in practical applications.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): • 4 Quizzes (3 Quizzes Compulsory) – 15 marks (5 marks for each) • 1 Android App Development Assignment from the provided list of applications -15 marks	30 (15 + 15)
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus:

Introduction to Android Development: Overview of Mobile App Development, Introduction to Android OS and its Architecture, Setting up the Android Development Environment (Android Studio),Understanding Android Project Structure, Basics of XML for UI Design **Android UI Components and Layouts :**View and View Groups, Common UI Widgets (Text View, Edit Text, Button, Image View,

etc.),Layouts: Linear Layout, Relative Layout, Constraint Layout, Recycler View and List View, Handling User Input and Event Listeners **Activities, Intents, and Navigation:** Activity Lifecycle and State Management, Implicit and Explicit Intents, Passing Data Between Activities, Fragments and Fragment Management, Navigation Components and Multi-Screen Applications **Data Persistence and Storage:** Shared Preferences for Data Storage, SQLite Databases in Android, Room Persistence Library, File Storage, Basics of Cloud Integration (Firebase Realtime Database)**Advanced Topics in Android Development:** Networking: Using APIs and JSON Parsing, **Background Tasks:** Async Task, Work Manager, and Services, Notifications and Broadcast Receivers, Integrating Sensors (e.g., GPS, Accelerometer),Basics of Jetpack Components.

Textbook:

1. Android Developers Official Documentation (developer.android.com).
2. Joseph Annuzzi Jr., Lauren Darcey, and Shane Conder, Android Application Development, Pearson Education, 2014.
3. B. Phillips, C. Stewart, B. Hardy, and K. Marsicano, Android Programming: The Big Nerd Ranch Guide, (4e), Big Nerd Ranch LLC, 2019.
4. Phil Dutson, Android Development Patterns: Best Practices for Professional Developers, (1e), Addison Wesley,2016.

Reference Book:

1. Dawn Griffiths and David Griffiths, Head First Android Development, (2e), Shroff/O'Reilly, 2017.

Online References:

1. www.youtube.com/@AndroidDevelopers
2. Kotlin Android Tutorial - GeeksforGeeks

G. Lecture Plan:

Lect. No.	Topics	Session Outcome	CO	Mode of Delivery	Mode of Assessing CO
1	Overview of Mobile App Development	Understand the mobile app ecosystem and its significance	CSE 3142.1	Lecture-based teaching-learning	CWS, MTE, ETE
2	Android OS and Architecture	Describe Android platform architecture and layers	CSE 3142.1	Lecture-based teaching-learning	CWS, MTE, ETE
3	Setting up Android Studio	Set up Android development environment	CSE 3142.1	Technology based learning	CWS
4	Android Project Structure	Identify components of Android project	CSE 3142.1	Lecture-based teaching-learning	CWS, MTE
5	Basics of XML for UI Design	Understand UI design using XML	CSE 3142.2	Technology based learning	CWS

6	Views and ViewGroups	Differentiate between Views and ViewGroups	CSE 3142.2	Technology based learning	CWS
7	Common UI Widgets	Use widgets like TextView, Button, EditText	CSE 3142.2	Technology based learning	CWS, Mini Project
8	Layouts: Linear, Relative	Design interfaces using Linear and Relative Layouts	CSE 3142.2	Technology based learning	CWS
9	Constraint Layout	Build responsive UI using Constraint Layout	CSE 3142.2	Technology based learning	CWS
10	ListView and RecyclerView	Display dynamic lists using adapters	CSE 3142.2	Technology based learning	CWS, MTE
Quiz-1					
11	Handling User Input	Implement input validation and response handling	CSE 3142.2	Technology based learning	CWS
12	Event Listeners	Use different types of event listeners	CSE 3142.2	Technology based learning	CWS
13	Activity Lifecycle	Describe activity lifecycle and state changes	CSE 3142.3	Lecture-based teaching-learning	MTE
14	Intents: Implicit & Explicit	Navigate between screens using intents	CSE 3142.3	Technology based learning	CWS, MTE
15	Passing Data Between Activities	Implement intent extras and bundles	CSE 3142.3	Technology based learning	CWS, MTE
16	Fragments and Fragment Management	Use fragments and back stack navigation	CSE 3142.3	Technology based learning	CWS, MTE
17	Navigation Components	Build multi-screen apps with navigation components	CSE 3142.3	Technology based learning	Mini Project
18	Shared Preferences	Store and retrieve simple data	CSE 3142.4	Technology based learning	CWS, MTE
19	SQLite Databases	Manage local storage using SQLite	CSE 3142.4	Technology based learning	CWS, MTE
20	Room Persistence Library	Implement Room for structured database access	CSE 3142.4	Technology based learning	CWS, MTE
Quiz -2					
21	File Storage	Handle internal and external storage	CSE 3142.4	Lecture-based teaching-learning	CWS, MTE

22	Firebase Realtime Database	Integrate cloud storage and sync	CSE 3142.4	Technology based learning	Mini Project
23	APIs and JSON Parsing	Fetch and parse web data	CSE 3142.5	Technology based learning	CWS, MTE
MID Term Evaluation					
24	Async Task and Work Manager	Perform background tasks efficiently	CSE 3142.5	Lecture-based teaching-learning	CWS, ETE
25	Services and Notifications	Implement background services and notifications	CSE 3142.5	Technology based learning	CWS, ETE
26	Broadcast Receivers	Handle system-wide events	CSE 3142.5	Technology based learning	CWS, ETE
27	Integrating Sensors (GPS, Accelerometer)	Access and use device sensors	CSE 3142.5	Technology based learning	CWS, ETE
28	Jetpack Components Overview	Understand Jetpack architecture and tools	CSE 3142.5	Lecture-based teaching-learning	CWS, ETE
29	Practical Review: UI & Navigation	Reinforce learning on UI & navigation	CSE 3142.5	Flipped Classroom	CWS, ETE
QUIZ-3					
30	Mini Project Design Discussion	Plan and define the mini project scope	CSE 3142.5	Flipped Classroom	CWS, ETE
31	Error Handling and Debugging in Android	Use tools for error detection and debugging	CSE 3142.5	Technology based learning	CWS, ETE
32	Performance Optimization	Optimize app performance and resource usage	CSE 3142.5	Lecture-based teaching-learning	CWS, ETE
33	Permissions and Security in Android	Understand Android permissions and security model	CSE 3142.5	Lecture-based teaching-learning	CWS, ETE
34	Publishing and Packaging Apps	Learn to package and deploy Android apps	CSE 3142.5	Technology based learning	ETE
QUIZ-4					
35	Mini Project Presentations I	Demonstrate working of developed apps	3142.5	Flipped Classroom	ETE
36	Mini Project Presentations II	Demonstrate working of developed apps	3142.5	Flipped Classroom	ETE
END TERM Evaluation					

Pedagogical Approaches:

I. Lecture based teaching-learning
II. Technology based learning
III. Learning through problem-solving
IV. Flipped Classroom

H. Learner centric activities:

Sample activities at low-difficulty level:

- One-on-one remedial sessions or tutorials
- Quizzes with immediate feedback for concept reinforcement
- Step-by-step problem-solving exercises
- Use of visual aids or concept maps for better comprehension

Sample activities at high-difficulty level:

- Assignments involving real-life application or case studies
- Mini research projects or presentations
- List out the online study materials on the subjects from top experts in the domain
- Outside Class Engagement Activities:

Sample Outside Class Engagement Activities:

- Participating in academic clubs, technical competitions, or seminars
- Collaborating with faculty or peers on research ideas or innovative prototypes
- Attending faculty-guided doubt-clearing or reinforcement sessions
- Watching recommended video lectures, tutorials, or educational webinars
- Visiting laboratories or project labs for extended hands-on practice

I. Course Articulation Matrix:

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CSE 3142.1	Recall the architecture of the Android operating system and the key components of the Android development environment. [L1; Remember]	3	2	-	-	-	-	2	3	2	-	-	-	-	2	3
CSE 3142.2	Explain the use of XML for UI design and demonstrate the implementation of various Android UI components and layouts. [L2; Understand]	2	-	3	3	-	-	-	2	-	3	3	-	-	-	2
CSE 3142.3	Develop Android applications using activities, intents, and fragments to create multi-screen applications and manage user interactions. [L3; Apply]	2	-	3	3	2	-	-	2	-	3	3	2	-	-	2
CSE 3124.4	Build data persistence techniques such as Shared Preferences and SQLite databases. [L3; Apply]	3	-	-	3	-	2	2	3	-	-	3	-	2	2	3
CSE3 142.5.	Analyze and integrate advanced features such as networking APIs, background tasks, notifications, and sensors to enhance the functionality of Android applications. [L4; Analyze]	3	-	-	3	-	2	2	3	-	-	3	-	2	2	3

Faculty Consultation Time:

1. Dr. Usha Jain → Wednesday (4PM- 5pm)
2. Dr. Surbhi Sharma → Monday (4PM- 5pm)
3. Ms. Kirti Paliwal → Monday (11AM- 12pm)